Zhang *et al. Cardiovascular Diabetology* (2026) 25:44
https://doi.org/10.1186/s12933-025-03041-8

Cardiovascular Diabetology

RESEARCH Open Access

The shared genetic architecture underlying the autoimmune and cardiovascular disease: a multivariate genome-wide analysis

Jie Zhang[1,7†], Xinyu Fang[2,3,4†], Qianling Ye[5†], Xianyong Yin[6*] and Dongqing Ye[1,2,3,7*]

Abstract

Background Epidemiological studies have established that autoimmune diseases (AD) increase cardiovascular disease (CVD) risk. We aimed to elucidate their shared genetic architecture and clinical implications.

Methods We conducted a cross-trait multivariate genome-wide association study (GWAS) for three autoimmune diseases (type 1 diabetes, systemic lupus erythematosus, and rheumatoid arthritis) and four cardiovascular diseases (coronary artery disease, heart failure, stroke, and peripheral artery disease). We performed both genome-wide and locus-based analysis including fine-mapping, functional annotation, enrichment analyses, transcriptome-wide association studies (TWAS), and proteome-wide and drug-repurposing Mendelian randomization (MR). We constructed a polygenic risk score (PRS) and explored its pleiotropic effects beyond AD and CVD in the UK Biobank population.

Results In the multivariate GWAS, we identified 259 genome-wide significant association signals that were enriched primarily in arterial tissues and lipid metabolism pathways. Through convergent evidence from TWAS and MR analyses, we prioritized 15 therapeutic targets including *TGFB1* and *IL6R*, and identified histone deacetylase inhibitors as candidate drugs. The polygenic risk score showed discriminative ability in cardiovascular risk stratification among autoimmune patients, with individuals in the highest PRS decile exhibiting significantly elevated CVD risk compared to those in the 2nd-9th deciles (hazard ratios: 1.60 [95% CI: 1.19–2.15] for type 1 diabetes, 2.28 [95% CI: 1.14–4.56] for systemic lupus erythematosus, and 2.33 [95% CI: 1.94–2.80] for rheumatoid arthritis). Additionally, the PRS revealed pleiotropic associations with risk of various health conditions, including polyneuropathy, chronic renal failure, and depressive episodes.

Conclusion Our study unveils the shared genetic architecture between autoimmune and cardiovascular diseases, providing insights for therapeutic development and risk stratification.

[†]Jie Zhang, Xinyu Fang and Qianling Ye have contributed equally to this work.

*Correspondence:
Xianyong Yin
xianyongyin@njmu.edu.cn
Dongqing Ye
ydqph@aust.edu.cn

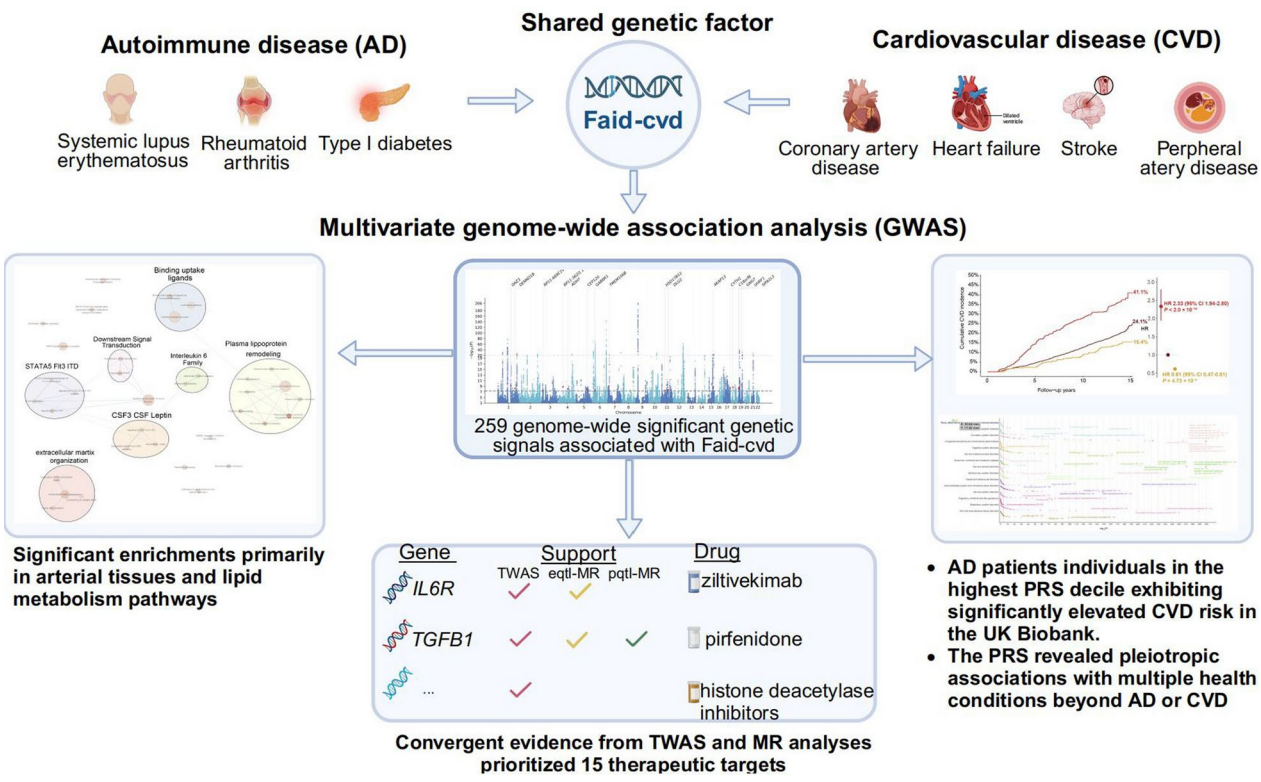
Full list of author information is available at the end of the article

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Keywords Autoimmune disease, Cardiovascular disease, Shared genetic architecture, Genome-wide association studies

Graphical abstract



Research Insights

What is currently known about this topic?

- Autoimmune diseases (AD) consistently associate with elevated cardiovascular disease (CVD) risk beyond traditional risk factors.
- Previous genome-wide association studies have revealed polygenic architecture of both AD and CVD separately.
- Limited understanding exists of the shared genetic mechanisms underlying AD-CVD comorbidity at the multi-traits level.

What is the key research question?

- What is the shared genetic architecture between autoimmune diseases and cardiovascular diseases, and can these insights inform therapeutic strategies and risk stratification?

What is new?

- First multivariate GWAS identified 259 shared genetic variants and 19 novel loci underlying AD-CVD comorbidity using genomic structural equation modeling.
- Prioritized 15 therapeutic targets including *TGFB1* and *IL6R* through convergent evidence from multiple analytical approaches.
- Constructed the polygenic risk score of Faid_cvd (shared genetic factor between autoimmune and cardiovascular diseases) that effectively stratifies cardiovascular risk in autoimmune patients, with 1.6–2.33 fold increased risk in high genetic risk groups.

How might this study influence clinical practice?

- Findings enable personalized cardiovascular risk stratification in autoimmune patients and identify potential therapeutic targets for dual AD-CVD treatment approaches.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Introduction

Autoimmune diseases (AD) represent a heterogeneous group of disorders characterised by immune system dysregulation, resulting in loss of self-tolerance and subsequent tissue damage [1]. Although ADs present with diverse clinical manifestations, they consistently associate with elevated cardiovascular disease (CVD) morbidity [2]. Notably, traditional cardiovascular risk factors fail to fully account for the increased CVD risk observed in autoimmune patients [2]. Understanding the genetic architecture underlying the AD-CVD comorbidity is therefore crucial for identifying shared pathogenic mechanisms and developing targeted therapeutic strategies [3].

Over recent decades, genome-wide association studies (GWASs) have revealed the polygenic architecture of both AD [4] and CVD [5], demonstrating extensive shared genetic effects within their respective disease clusters. Mounting evidence also supports genetic links between specific autoimmune disorders and cardiovascular manifestations. For instance, studies have demonstrated that shared pathogenic mechanisms between rheumatoid arthritis (RA) and CVD arise from common genetic components [6]. However, the shared genetic basis of AD-CVD comorbidity likely involves a complex interplay of multiple genetic variants with modest effect sizes distributed across the genome, rather than being driven by a few major loci. Such genetic complexity necessitates the implementation of cross-trait multivariate genomic approaches to better delineate their shared aetiological landscape.

In this study, we focused on three common autoimmune disorders (RA, systemic lupus erythematosus [SLE], and type 1 diabetes [T1D]), selected based on their established high genetic correlation [7], rather than encompassing the full spectrum of heterogeneous autoimmune conditions. Our analytical framework comprised several sequential steps (Fig. 1). Initially, we examined pairwise genetic correlations between these three ADs and four cardiovascular diseases (coronary artery disease [CAD], heart failure [HF], stroke, and peripheral artery disease [PAD]). Subsequently, we employed genomic structural equation modelling (genomic SEM) [8] to characterize model shared genetic architecture—termed Faid_cvd as the latent factor of AD-CVD comorbidity—using multivariate GWAS summary statistics across the selected ADs and CVDs, followed by multivariate GWAS of Faid_cvd and post-GWAS analyses. We then leveraged proteome-wide and drug repositioning Mendelian randomization approaches, combined with computational drug repurposing analysis, to identify potential therapeutic targets. Additionally, we constructed polygenic risk scores representing shared genetic liability to assess their performance in population-level risk stratification in the UK Biobank cohort. Finally, we conducted phenome-wide association studies to explore pleiotropic associations beyond AD- or CVD-related phenotypes.

Methods

GWAS summary statistics sources

All data sources utilized in this study were derived from publicly available GWAS summary statistics conducted in European ancestry populations using genome-wide genotyping arrays. These methodological criteria were essential for ensuring compatibility with genomic SEM analysis, which requires accurate LD score regression (LDSC) estimation. We have investigated three autoimmune diseases: RA (cases $n = 22{,}350$, $N = 97{,}173$) [9]; T1D (cases $n = 18{,}942$, $N = 520{,}580$) [10]; SLE (cases $n = 5201$, $N = 14{,}267$) [11], and five cardiovascular diseases: CAD (cases $n = 181{,}523$, $N = 1{,}165{,}720$) [12]; HF (cases $n = 47{,}309$, $N = 977{,}323$) [13]; Stroke (cases $n = 73{,}652$, $N = 1{,}308{,}460$) [14]; PAD (cases $n = 12{,}086$, $N = 461{,}639$) [15] (Supplementary Table S1). Prior to our analyses, we performed a quality control of the GWAS summary statistics, related details are provided in Supplementary Methods. Genetic correlations between traits were estimated using LDSC [16] (excluding HLA region) and LAVA [17] for genome-wide and local genetic correlations, respectively, with significance assessed after multiple testing correction (Benjamini–Hochberg method). (More details on genetic correlation analyses see Supplementary Methods).

Multivariate GWAS of AD-CVD using Genomic SEM

Genomic SEM analysis [8] was conducted in two stages to uncover shared variance components across AD and CVD. Stage 1 involved exploratory factor analysis to inform subsequent confirmatory analysis. A two-factor model (Faid and Fcvd) was found optimal, with Faid loading onto autoimmune diseases (RA, SLE, T1D) and Fcvd loading onto cardiovascular diseases (CAD, HF, stroke, PAD), all factor loadings (β) exceeding 0.4. Model fit was assessed using comparative fit index (CFI) and standardised root mean square residual (SRMR) parameters, with thresholds of CFI > 0.95, SRMR < 0.04, and standardized factor loadings $\beta > 0.3$. A higher-order factor (Faid_cvd) incorporating both Faid and Fcvd was modeled, with Faid loading fixed at 1 for model identification (Supplementary Table S2). In Stage 2, we estimated SNP-specific effects for the Faid_cvd factor using 'userGWAS' function with default parameters from the weighted least squares (WLS) estimation method in genomic SEM , analyzing 5,375,214 SNPs shared across all AD and CVD GWAS. Heterogeneity was assessed using Q_{SNP} statistics comparing common and independent pathway models (Supplementary Fig. S1). Significant Q_{SNP} ($P < 5 \times 10^{-8}$) indicated SNP effects better explained by specific

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

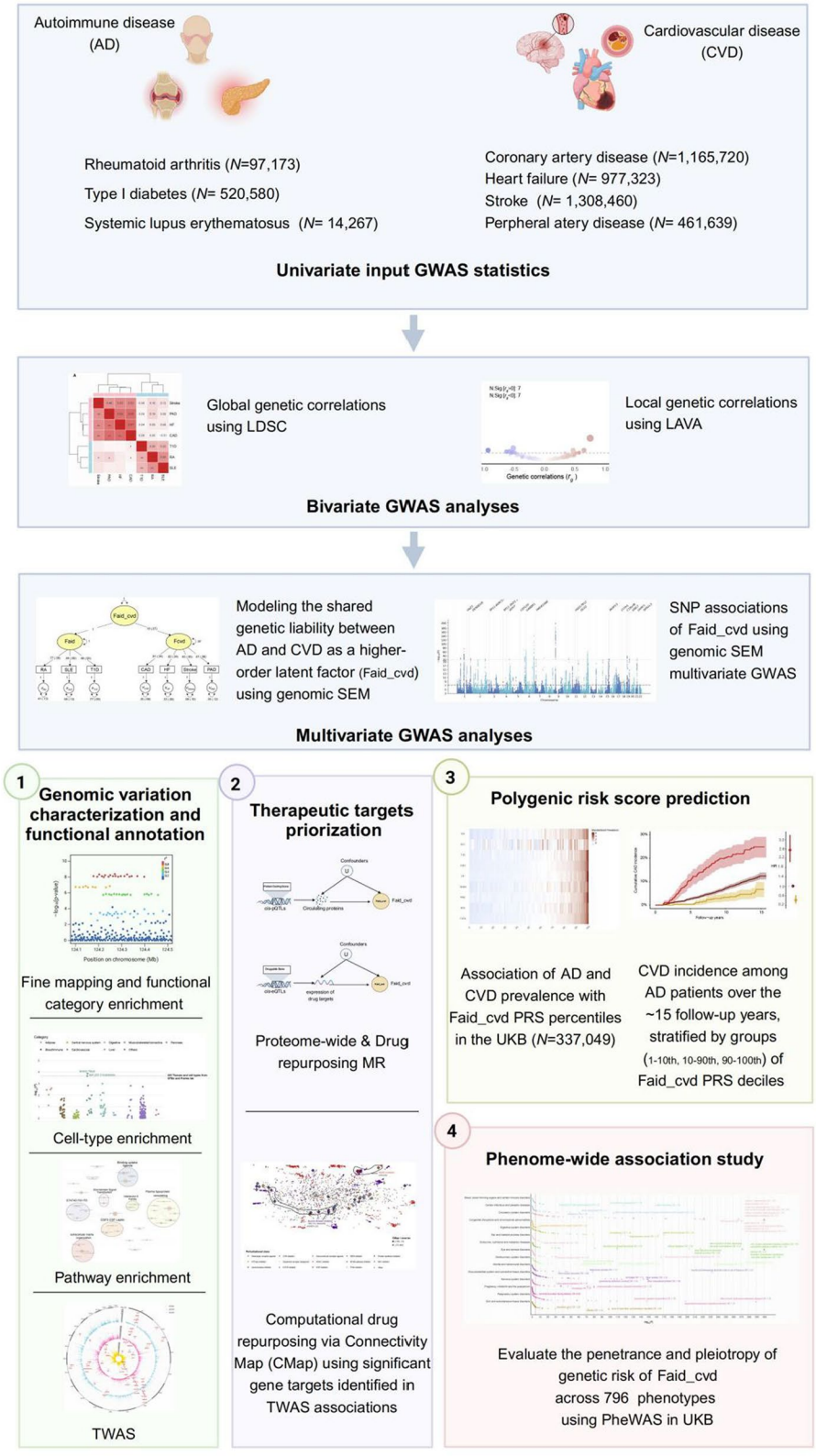


Fig. 1 An overview of this study's data sources, analytical flow and methodology. Created with BioRender.com

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

pathways independent of the shared Faid_cvd factor. (more details in Supplementary Methods).

Defining genomic loci, determining novel variants and fine-mapping

We employed the FUMA [18] v.1.3.5e to identify genomic loci and lead/sentinel SNPs associated with Faid_cvd. The lead SNPs were identified through a two-step clumping procedure: the independent significant SNPs were first defined by clumping SNPs with a $P < 0.05$ and $R^2 \geq 0.6$; the lead SNPs were defined by applying a second round of clumping with a $P < 5 \times 10^{-8}$ and $R^2 \geq 0.1$. The identified lead SNPs were subjected to GCTA-COJO [19] v.1.94.1 and determined whether they were conditionally and jointly associated SNPs through a stepwise model selection procedure with default parameters ($P < 5 \times 10^{-8}$, window of 10 Mb, and collinearity < 0.9). Loci were considered novel if they were > 1 Mb from loci identified in the component GWAS of Faid_cvd. We then employed SuSIE [20] and FINEMAP [21] to identify the most plausible causal variants associated with Faid_cvd. We analyzed a 250 kb window around each lead SNP in the Faid_cvd genomic loci, using a probability threshold of 0.95 to define credible sets of potentially causal variants (see Supplementary Methods for more information on fine-mapping).

SNP heritability, cell- and tissue-specific, gene-set and disease ontology enrichment analyses

We conducted a series of analyses to investigate the functional relevance and biological mechanisms underlying the shared genetic basis between AD and CVD (Faid_cvd). SNP heritability was partitioned across 53 functional annotations using S-LDSC [22]. Cell- and tissue-specific enrichment analyses were conducted using LDSC-SEG [23], testing heritability enrichment in genes exhibiting high cell- or tissue-specific expression, comprising: 205 tissues and cell types from the Genotype-Tissue Expression (GTEx) project and Franke laboratory datasets; 489 tissue-specific chromatin-based annotations from the Roadmap Epigenomics and 292 immune cell-types from ImmGen gene expression data. MAGMA gene-based and gene-set analyses mapped SNPs to 18,033 protein-coding genes, followed by FUMA 'GENE-2FUNC' analysis [24] using MSigDB gene sets. Mendelian disease genes and associated pathways enrichment was evaluated using MendelVar [25]. More details in Supplementary Methods.

Transcriptome-wide association analyses

We then conducted a transcriptome-wide association study (TWAS) to prioritize genes associated with Faid_cvd using the FUSION method [26]. We utilized TWAS weights for 37,920 precomputed expression quantitative trait loci features [27] from GTEx v.8, of which our Faid_cvd had sufficient variants to analyze 37,917 genes, surpassing Bonferroni correction for multiple comparisons ($P < 1.38 \times 10^{-6}$) were subjected to additional analysis. We considered TWAS-significant genes that were with high probability of being causal (FOCUS [28] posterior inclusion probability [PIP] > 0.5). Using the Connectivity Map (CMap) algorithm [29], we analyzed expression profiles of TWAS-significant genes to identify drugs capable of reversing these disease-associated expression patterns, with more negative τ scores indicating stronger drug repurposing potential. More details in Supplementary Methods.

Proteome-wide and drug-repurposing Mendelian randomization

Genetic instrument selection and validation for circulating protein levels have been previously detailed [30]. In brief, we employed high-confidence (tier 1) genetic instruments identified from independent ($R^2 < 0.001$) *cis*-pQTLs associated with circulating protein levels (GWAS $P \leq 5 \times 10^{-8}$), which passed specificity and consistency tests [30]. From the 738 circulating proteins with the validated genetic instruments, we harmonized these genetic instruments with our Faid_cvd GWAS summary statistics by matching SNP identifiers, chromosomal positions, and alleles. After harmonization, 494 proteins (66.9%) had complete genetic variant information available in both the protein GWAS and Faid_cvd GWAS, enabling subsequent proteome-wide MR analysis. The proteome-wide Mendelian randomization was performed using the Wald ratio method for each of these 494 genetic instruments.

For drug-repurposing genetic instrument selection [31], genes encoding targets of approved or clinical-stage therapeutics were identified using ChEMBL v.26. Conditionallyindependent *cis*-eQTLs for each gene were identified from GTEx v8. When multiple instruments from different GTEx v8 tissues were available, each tissue was tested independently. Two-sample MR estimated the effect of transcript levels of each drug target on Faid_cvd. Inverse-variance weighting combined effects for multiple genetic instruments per transcript. The false discovery rate (FDR) was controlled at 5% to account for multiple testing in above MR analyses.

Polygenic risk score of Faid_cvd and Phenome-wide association analysis in UK biobank

To construct a polygenic risk score (PRS) representing the Faid_cvd—which captures shared genetic liability to AD-CVD comorbidity—we extracted genetic variants from UK Biobank imputed genotyping data [32] (Details and quality control see Supplementary Methods). We selected SNPs based on the following criteria: (1) SNPs

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

significantly associated with Faid_cvd (we tested a series of *P*-value thresholds: 5×10^{-8}, 1×10^{-7}, 1×10^{-6}, 1×10^{-5}, 0.01, 0.05); (2) SNPs with minor allele frequencies > 0.01; and (3) no insertion or deletion or ambiguous SNPs. Clump-based linkage disequilibrium pruning was performed with a physical distance threshold of 250 kb and an LD threshold (R^2) < 0.01. Then we calculated the PRSs using PRSice-2 [33], as the following formula:

$$PRS_j = \frac{\sum_{\mathrm{i}}(S_j \times G_{ij}) - \mathrm{Mean}(PRS)}{\mathrm{SD}\ (PRS)}$$

We applied the genetic additive model, *S* is the summary statistic for the effective allele and *G* is the number of the effective allele observed, *i* is i_{th} of SNP, *j* is j_{th} of individual).

We ascertained the diagnosis of AD and CVD in the electronic health linkage of UK biobank. We then used logistic regression model to assess the associations of Faid_cvd PRSs with the above AD and CVD diagnosis, and adjusted for age, sex, assessment centers, and the top ten genetic principal components (PCs). Next, we calculated the phenotypic variance explained (Nagelkerke's R^2) of individual AD and CVD explained by Faid_cvd PRS, and effect size estimate (β) per unit of PRS. We applied a Cox proportional hazard model to estimate hazard ratios (HRs) and absolute risk of developing CVDs among individuals already diagnosed with ADs, which were stratified by three Faid_cvd PRS groups (first, second to ninth, and tenth deciles). We next conducted phenome-wide scan to test the pleiotropic associations of the Faid_cvd PRS with the 768 categorical disease phenotypes (defined as cases/non-cases, phenotypes < 200 cases have been prior excluded). (more details in Supplementary Methods).

Results

Pairwise genetic correlations

Linkage disequilibrium score regression revealed significant genetic correlations among ADs or CVDs. Of the 12 AD-CVD pairs examined, only RA showed significant genetic correlations with stroke ($r_g = 0.16$; $FDR < 0.05$) and PAD ($r_g = 0.19$; $FDR < 0.05$). Nominally significant genetic correlations were observed between RA and CAD ($r_g = 0.05$, $P = 0.044$), and between T1D and CAD ($r_g = 0.08$, $P = 0.034$) (Fig. 2a and Supplementary Table S3).

Local genetic correlations were estimated within 2495 LD-independent blocks using LAVA (Fig. 2b). We identified 126 significant local genetic correlations across 109 LD-independent blocks ($FDR < 0.05$) (Supplementary Fig. S2 and Supplementary Table S4). Three LD-independent blocks exhibited positive local genetic correlations across multiple (≥ 3) AD-CVD pairs. For instance, the 12q24.11–13 locus exhibited positive local genetic correlation between five disease pairs: T1D-CAD ($r_g = 0.75$), T1D-PAD ($r_g = 0.44$), T1D-stroke ($r_g = 0.74$), RA-CAD ($r_g = 0.44$), and RA-PAD ($r_g = 0.65$) (Fig. 2c). In this region, the genes *SH2B3* and *ATXN2* have been associated with disease susceptibilities of both autoimmune disorders [34] and vascular pathology [35].

However, the observed low to moderate global genetic correlations and sparse distribution of significant local genetic correlations provide limited insight into the genetic architecture underlying AD-CVD comorbidity, especially given the polygenic nature of these disorders [6–8].

Multivariate genetic factor model and multivariate GWAS

To model shared genetic factors underlying AD and CVD, we constructed a hierarchical factor model using genomic SEM [8], with robust model fit (CFI = 0.994; SRMR = 0.0377) (Supplementary Table S2): two primary latent factors (Faid/Fcvd) characterizing the common genetic liability of ADs/CVDs and a higher-order latent factor (Faid_cvd) capturing the shared genetic components between ADs and CVDs (Fig. 3a).

We assessed the associations of 5,375,214 SNPs shared among all the AD and CVD GWAS with Faid_cvd. The N_{eff} of the Faid_cvd was estimated to be 314,248 and the SNP-based heritability (h^2) was estimated to be 0.12. We identified 269 lead SNPs within 169 independent loci at genome-wide significance $P < 5 \times 10^{-8}$, of which 259 conditionally and jointly associated (COJO) Faid_cvd SNPs were further ascertained (Supplementary Tables S5 and S6). Among them, nine variants were non-synonymous, and eight variants had a CADD score of ≥ 12.37 (indicating these variants are possibly have deleterious effects). Moreover, 19 of these have not been reported in loci from previous AD or CVD GWASs (Fig. 3b and Table 1).

Among the 259 COJO Faid_cvd SNPs, only 11.2% ($n = 29$) were genome-wide significant in the heterogeneity test of Q_{SNP} (Supplementary Table S5). Among COJO Faid_cvd SNPs, 23.2% ($n = 60$) showed complete directional concordance across all components, while 53.7% ($n = 139$) demonstrated consistency in five or six Faid_cvd components (Supplementary Table S6). These results indicated that the identified COJO Faid_cvd SNPs exhibit consistent horizontal pleiotropic effects across Faid_cvd components through shared genetic liability (Faid_cvd factor).

Fine-mapping identifies putative causal variants

To identify putative causal variants within the 169 Faid_cvd loci, we performed a fine-mapping analysis using both SuSiE [20] and FINEMAP [21]. SuSiE identified 19 variants in 95% credible sets ($CS_{95\%}$), including 15 single-variant credible sets. FINEMAP yielded 31 $CS_{95\%}$, with 22 single-variant credible sets (Supplementary Table S7). We identified six variants as high-confidence causal

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

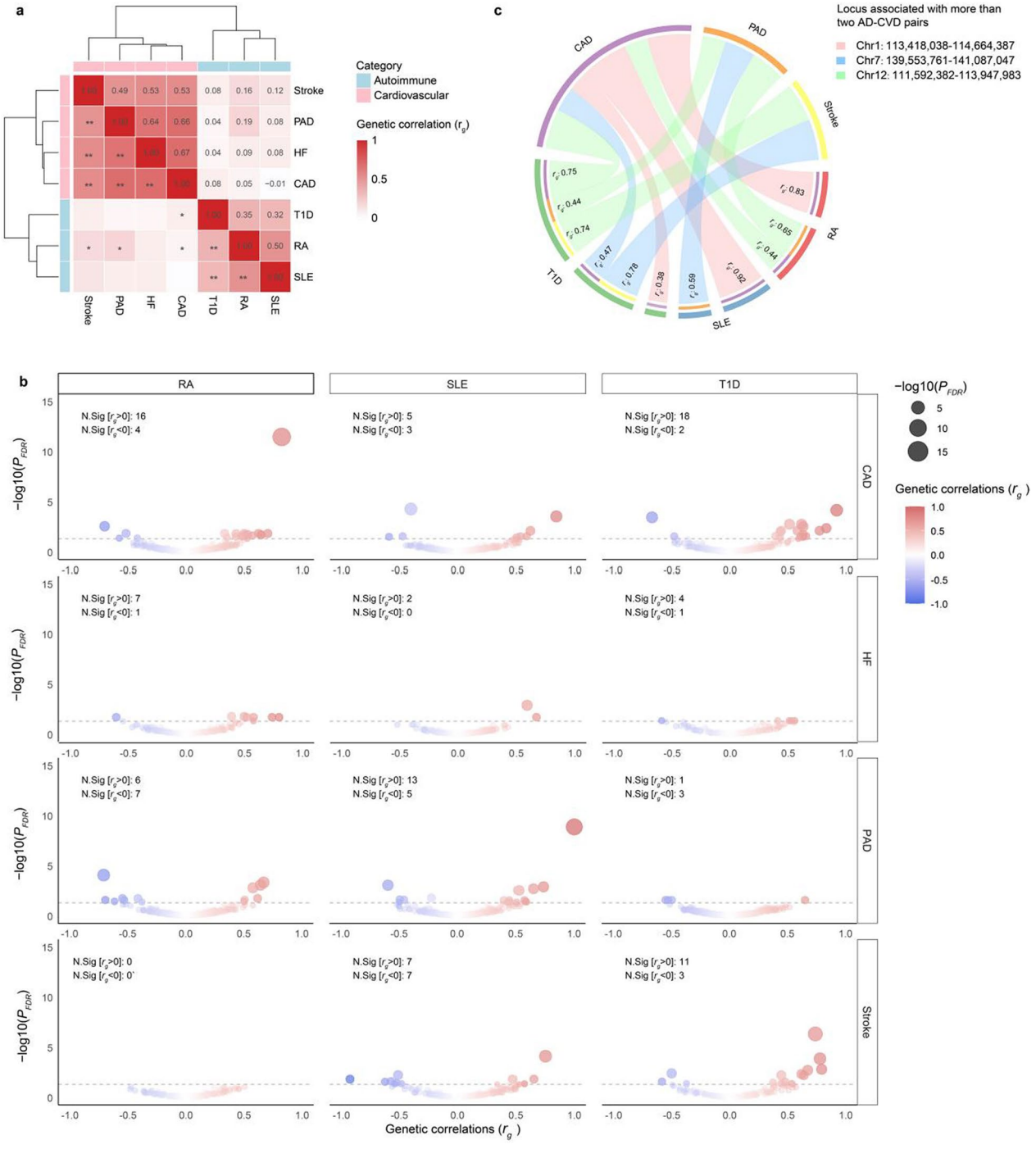


Fig. 2 Genomic and local genetic correlations between AD and CVD. **a** Matrix of bivariate genetic-correlation estimates, where the off-diagonal elements correspond to genetic correlations. Estimates that are non-significant are left blank. * denotes nominal $p < 0.05$ and ** denotes nominal $p < 0.01$. **b** Volcano plots based on LAVA results showing genomic lociwith the local genetic correlation between ADs and each of the CVDs (*x* axis) and the corresponding *P* value (*y* axis). Loci exceeding the horizontal line are significant at *FDR* < 0.05 (Benjamini-Hochberg-adjusted *P* value). Multiple testing was performed separately for each trait over all considered loci. **c,** locus where significant lcoal genetic correlations across multiple (≥ 3) AD-CVD pairs

variants, consistently supported by both FINEMAP and SuSiE (Supplementary Table S7 and Supplementary Fig. S3–S8). Among these six putative causal variants, four variants were exclusively contained within their respective single-variant 95% credible sets: rs11591147 (missense variant, *PCSK9*), rs55730499 (intronic variant, *LPA*), rs9515201 (intronic variant, *COL4A2*), and rs7412 (missense variant, *APOE*) (Supplementary Fig. S9). These variants have known implications in both AD and CVD pathogenesis, as exemplified by rs11591147, a

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

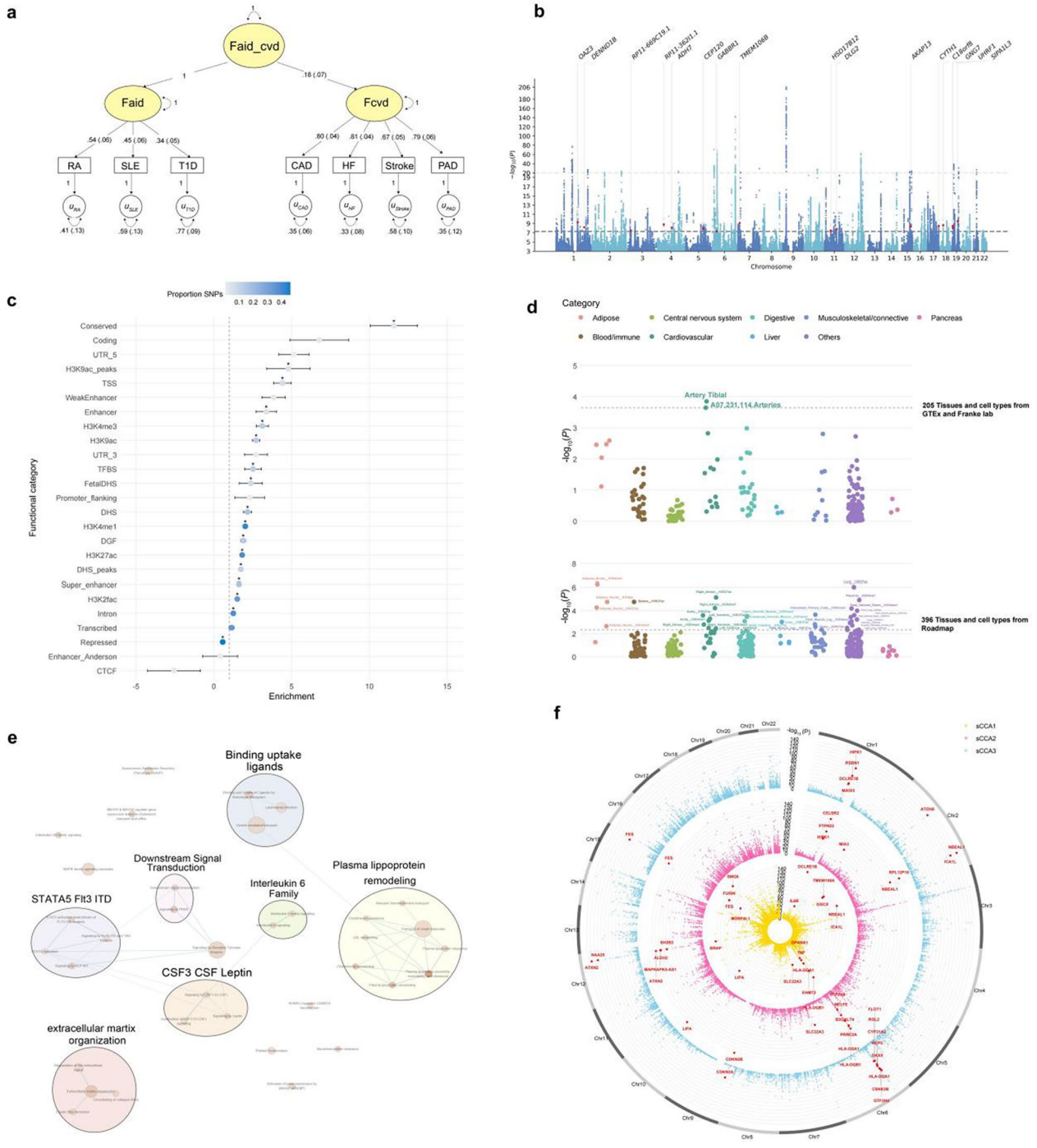


Fig. 3 (See legend on next page.)

loss-of-function mutation in *PCSK9* previously linked to coronary disease [36] and T1D [37].

Functional annotation, tissue and cell-specific and pathway enrichment analyses

We performed partitioned SNP heritability analysis using S-LDSC [22] regression across 53 annotations to investigate functional category-specific heritability enrichment. Significant enrichment was observed in 31 annotations after Bonferroni correction ($P < 9.43 \times 10^{-4}$). The strongest enrichments of SNP heritability were detected in conserved genomic regions (enrichment = 11.6; s.e. = 1.5), transcription start site (TSS) regions (enrichment = 8.3; s.e. = 2.0), and H3K9ac peaks of genes (enrichment = 6.4; s.e. = 1.5) (Fig. 3c and Supplementary Table S8).

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

(See figure on previous page.)

Fig. 3 Share genetic architecture of AD-CVD modeled with genomic SEM and the subsequent multivariate GWAS and post-GWAS analyses. **a** The latent factor model as specified in genomic SEM with the 'observed' phenotypes in rectangles and the latent variables in circles. Factor loadings (standardized with respect to the genetic variance of the traits) are given in the plot. **b** Latent AD-CVD factor (Faid_cvd) GWAS results. The *x* axis shows the genomic position and the *y* axis shows statistical significance as $-\log_{10}P$. The 19 novel loci are displayed. **c** Heritability enrichment for functional SNP categories. The x axis represents functional categories, and the y axis represents enrichment. Error bars represent the 95% CIs, calculated as enrichment ± 1.96 × s.e. Asterisks represent significant enrichment after Bonferroni correction (uncorrected *P* from two-sided enrichment test < 9.43×10^{-4}). **d** Tissue or cell-specific enrichment analysis using LDSC-SEG. The *x* axis represents tissue categories, and the *y* axis represents the uncorrected $-\log_{10}(P)$ from one-sided *z* test for enrichment in specific tissues. The black dashed line represents the significance threshold of *FDR* < 0.05. The up panel showed the enrichment results from gene expression data of 205 tissue and cell type from GTEx and Franke lab ; The up panel showed the enrichment results from activating histone marks data of 396 tissue and cell type from Roadmap **e** The significant enrichment Reactome pathways (*FDR* < 0.05). Related gene-sets were grouped into network clusters using "Enrichment Map" in Cytoscape. A node in the Enrichment Map represents a gene set. Node size represents the gene set size and line thickness shows the degree of overlap (shared genes) between the two gene sets it connects. **f** Circos plot of TWAS results for Faid_cvd. The y axis represents the uncorrected $-\log_{10}(P)$. Bonferroni-corrected significance threshold of $P = 1.38 \times 10^{-6}$ (0.05/37,917 genes) was set. The points highlighted with red color present high-confidence genes ($P < 1.38 \times 10^{-6}$ and posterior inclusion probability > 0.5), here only show the top 20 under each cross-tissue eQTL weights (sCCA1, sCCA2 and sCCA3) . Chr, chromosome; CTCF, CCCTC-binding factor; DGF, digital genomic footprint; DHS, DNase I hypersensitivity site; TFBS, transcription factor binding site; TSS, transcription start site; UTR, untranslated region

Using LDSC-SEG [23] analysis, we examined Faid_cvd genetic enrichment in cell- and tissue-specific gene expression and histone marks. We identified significant enrichment in arterial gene expression ($P = 2.3 \times 10^{-4}$) and 31 activating histone marks, such as adipose nuclei H3K4me1 ($P = 5.6 \times 10^{-7}$) and osteoblast primary cells H3K4me3 ($P = 2.0 \times 10^{-4}$) (Fig. 3d, Supplementary Tables S9 and S10). Enrichment analysis of 292 immune cell types showed nominal associations with several T cell, stromal cell, and myeloid cell subsets, though none remained significant after multiple testing correction (Supplementary Table S11).

Multimarker analysis of genomic annotation (MAGMA) [24] gene set analysis identified multiple pathways that were associated significantly with Faid_cvd, including plasma lipoprotein remodeling, interleukin-6 family signaling, STAT5/Flt3/ITD signaling cascade, CSF3/CSF/Leptin-mediated signaling, extracellular matrix organization, ligand-binding and uptake mechanisms, and downstream signal transduction (Fig. 3e and Supplementary Table S12). Additionally, pathway enrichment analysis using MendelVar [25] identified 55 Faid_cvd-related Mendelian diseases, spanning lipid metabolism disorders, immune system diseases, cardiovascular system diseases, and diabetes mellitus (Supplementary Table S13).

Transcriptome-wide association analysis

TWAS identified 310 gene expressions significantly associated with Faid_cvd ($P < 1.38 \times 10^{-6} = 0.05/37{,}917$ after Bonferroni correction) (Fig. 3f and Supplementary Table S14). The *DCLRE1B* at 1p13.2 exhibited the highest positive TWAS *Z*-score ($Z = 15.7$, PIP = 1), consistent with its previous identification as an independent locus shared between RA and CVD traits [6]. Additionally, *CDKN2B* at the established cardiovascular risk locus 9p21 exhibited the strongest negative TWAS *Z*-score ($Z = -25.7$, PIP = 1). This gene regulates TGFβ signaling and smooth muscle cell investment of hypoxic neovessels, and has also been associated with coronary artery calcium levels in SLE patients [38].

Identification of potential drug target and drug reposition

We identified significant associations for 22 out of 494 proteins that had available *cis*-pQTL genetic instruments. (Fig. 4a and Supplementary Table S15). These 22 proteins showed network interactions, particularly co-expression (Supplementary Fig. S10), and were enriched in pathways related to cell migration, epithelial cell proliferation, and leukocyte migration (Supplementary Table S16). Among them, 14 proteins were further replicated using *cis*-pQTLs from an independent proteome GWAS (Supplementary Table S17).

Expression levels of 148 unique targets of all 879 actionable therapeutic targets showed significant associations with Faid_cvd (Fig. 4b and Supplementary Table S18). Among the drug targets significantly associated with Faid_cvd, the 15 of 148 target genes were also found in our TWAS results (Fig. 4c). A high-priority candidate target for drug repurposing is Transforming Growth Factor Beta 1 (*TGFB1*), supported by multiple lines of evidence: TWAS ($Z = -5.84$, PIP = 0.90), *cis*-pQTLs MR (OR 0.94, $FDR = 3.96 \times 10^{-10}$), and drug repurposing MR (ORs 0.91–0.94, $FDR = 3.61 \times 10^{-10}$) (Fig. 4c), all these analyses converge on this target gene, and its approved drug pirfenidone, an antifibrotic agent with anti-inflammatory and antioxidant properties, which has demonstrated clinical utility in treating rheumatoid arthritis-associated interstitial lung disease and has shown promising efficacy in reducing myocardial fibrosis among patients with heart failure with preserved ejection fraction, as evidenced by the recent randomized phase II trials [39]. Another significant target identified was interleukin 6 receptor (*IL6R*) supported by TWAS ($Z = -8.45$, PIP = 1) and drug repurposing MR (ORs 0.91, $FDR = 5.47 \times 10^{-15}$) (Fig. 4c), which is targeted by the established rheumatoid arthritis drug sarilumab. Significant progress in cardiovascular outcomes research has recently been also

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

achieved with ziltivekimab—a novel IL-6 ligand monoclonal antibody specifically developed for atherosclerosis [40].

We performed computational drug repurposing analysis using Connectivity Map (CMap) [29] based on significant gene targets identified through TWAS associations. The analysis revealed several clinically relevant drug classes, notably histone deacetylase (HDAC) inhibitors and aurora kinase inhibitors (Fig. 4c). The identification of HDAC inhibitors is particularly noteworthy, as this drug class is emerging as a potential therapeutic approach for autoimmune and cardiovascular diseases [41], supporting the validity of our drug repurposing findings.

Polygenic risk scores improve the risk stratification of AD and CVD comorbidity

We constructed polygenic risk scores (PRS) using Faid_cvd summary statistics and examined their associations with AD and CVD diagnoses in 337,049 unrelated European ancestry participants in UK biobank. The Faid_cvd PRS demonstrated varying degrees of R^2 across these diseases (all $P < 1.1 \times 10^{-10}$; Fig. 5a and Supplementary Table S19): CAD exhibited the highest phenotypic variance explained ($R^2 = 0.106$), followed by HF, T1D and PAD ($R^2 = 0.0423$, 0.0332 and 0.0239, respectively). Relatively lower levels of explained variance were observed for stroke, RA and SLE ($R^2 = 0.0087$, 0.0087 and 0.0044, respectively). Correspondingly, we observed markedly increased odds ratios (OR) ranging from SLE diagnosis (OR 1.30, 95% CI 1.20–1.41) to CAD diagnosis (OR 2.36, 95% CI 2.33–2.39) per SD increasing in Faid_cvd PRS.

We observed the associations between Faid_cvd PRS and disease prevalence across AD and CVD, with population in the top PRS decile showing markedly higher disease prevalence (Fig. 5b). We also found that higher Faid_cvd PRS associated with increased risk overall CVD events in AD patients in the multivariable Cox regression models (Fig. 5c-e). For instance, RA patients in the top Faid_cvd PRS decile showed a significantly higher risk of incident CVD (HR 2.33, $P < 2.0 \times 10^{-16}$) compared to those in the middle range (2nd–9th decile). During the 15-year follow-up period, the cumulative CVD incidence in the top PRS decile was almost threefold higher than in the bottom decile (41.1% vs. 15.4%; Fig. 5c).

Phenome-wide scan reveals pleiotropic associations with extensive health conditions

Approximately half of phenotypes ($n = 305$) met the phenome-wide significance threshold ($P = 1 \times 10^{-5}$) after the phenome-wide scans. Beyond these anticipated associations with all constituent phenotypes (RA, SLE, T1D, CAD, HF, stroke, PAD) of Faid_cvd, we also found that PRS of Faid_cvd was associated with diverse disease categories, including genitourinary system disorders (chronic renal failure, OR 1.32, $P = 3.16 \times 10^{-305}$); mental and behavioural disorders (mental and behavioural disorders of tobacco use, OR 1.23, $P = 1.17 \times 10^{-233}$; depressive episode, OR 1.11, $P = 4.81 \times 10^{-79}$); nervous system disorders (polyneuropathy, OR 1.76, $P = 9.99 \times 10^{-81}$); and respiratory system disorders (chronic obstructive pulmonary disease, OR 1.32, $P = 2.32 \times 10^{-290}$) (Fig. 6 and Supplementary Table S20).

Discussion

In this work, we found that generally shared genetic basis between three ADs (RA, SLE and T1D) and four CVDs (CAD, HF, Stroke, PAD) cannot be completely explained by genetic correlations. Therefore, we modeled a cross-phenotype shared genetic factor (Faid_cvd) using genomic SEM to present common genetic liability of AD-CVD comorbidity. We next performed a multivariate GWAS for Faid_cvd with a resultant effective sample size of 314,248 participants and discovered 19 previously unreported risk loci, enrichment analyses from pathway, cell types and Mendelian diseases highlight the importance of lipid metabolism, immune inflammatory response (IL6 signaling), extracellular matrix remodeling and cell signaling transduction (like G-CSF, STAT5 and FLT3-TKD) mechanisms in AD-CVD comorbidity. Subsequently, we developed a polygenic risk score (PRS) to characterize the polygenic nature of Faid_cvd. This PRS demonstrated significant associations with both ADs and CVDs in UKB general population, and suggesting its utility in enhancing prediction of CVD incidence among individuals with ADs. Additionally, we provided evidence for pleiotropic associations with conditions such as polyneuropathy, chronic renal failure and depressive episodes. Lastly, we prioritized candidate therapeutic targets (e.g. *SWAP70*, *CSF1*, *TGFB1*,*IL6*, and *PCSK9*) via integration of pQTLs and eQTLs data. These findings substantially enhance our knowledge of the genetic basis of underlying the high comorbidity between the selected AD and CVD, and offer an opportunity to advance early public health intervention, precise risk prediction and facilitate novel drug development or drug repurposing strategies.

Our findings about genetic correlations between ADs and CVDs are concordant with previous studies [6, 42–44]. We identified significant local genetic correlations across multiple AD-CVD pairs. Notably, the genomic locus region on chromosome 12 (111,592,382–113,947,983), encompassing genes such as *SH2B3* and *ATXN2*, exhibited strong local genetic correlations between T1D, RA and the same set of CVDs (CAD, PAD, stroke), these candidate pleiotropic genes also were confirmed between ADs and CVDs in previous investigation [42]. The interplay between *SH2B3*-mediated

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Table 1 The 19 novel loci identified in the shared genetic factor of Faid_cvd

SNP	Location (chr:pos)	OA/EA	MAF	Beta (SE)	*P* value	Nearest mapped gene (SNP function)	Q_{SNP} *P* value	Previous GWAS associations (*P* value < 5×10^{-8})
rs1196456	1:151,737,349	A/T	0.14	−0.0203 (0.0032)	5.23E-10	*OAZ3:RP11-98D18.3 (ncRNA_intronic)*	0.17	Coronary artery disease, Myocardial infarction
rs4915551	1:197,508,901	A/G	0.23	−0.0177 (0.0030)	6.39E-09	*DENND1B (intronic)*	0.49	–
rs6801593	3:20,465,084	T/G	0.3171	0.0151(0.0027)	3.50E-08	*RP11-669C19.1(intergenic)*	0.4	Body mass index, Arm fat mass, Impedance of whole body
rs10516440	4:100,305,393	G/A	0.1183	−0.0212(0.0037)	7.47E-09	*ADH7(intergenic)*	0.37	Alcohol intake frequency, Low density lipoprotein cholesterol levels, Alcohol related diseases and deaths
rs12641981	4:45,179,883	T/C	0.4185	−0.0154(0.0025)	1.53E-09	*RP11-362I1.1(intergenic)*	0.16	Body mass index, Body fat percentage, Diabetes mellitus, Obesity, coronary artery disease, physical activity measurement, comparative body size at age 10
rs2081914	5:122,673,622	G/C	0.4751	0.0142(0.0025)	1.21E-08	*CEP120(intergenic)*	0.37	Low density lipoprotein cholesterol levels, Sex hormone-binding globulin, Heel bone mineral density
rs9469427	6:33,429,275	T/A	0.02883	−0.0657(0.0069)	1.38E-21	*ZBTB9(intergenic)*	**1.78E-15**	Hypothyroidism or myxedema, Rheumatoid arthritis, Lung function (FEV1/FVC), Methotrexate, Levothyroxine sodium, Asthma, Diabetes or endocrine disease, White blood cell count, Celiac disease, Neutrophil count, Monocyte percentage of white cells, Sex hormone-binding globulin
rs11509880	7:12,261,911	G/A	0.328	−0.0163(0.0027)	7.74E-10	*TMEM106B(intronic)*	0.61	Neurociticism, Coronary artery disease
rs2625389	11:43,611,196	T/A	0.3996	0.0143(0.0026)	3.20E-08	*HSD17B12(intergenic)*	0.15	Trunk fat mass, Whole body fat mass, menarche, Lung function (FVC), Sex hormone-binding globulin levels

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Table 1 (continued)

SNP	Location (chr:pos)	OA/EA	MAF	Beta (SE)	*P* value	Nearest mapped gene (SNP function)	Q_{SNP} *P* value	Previous GWAS associations (*P* value < 5×10^{-8})
rs108499	11:61,547,237	C/T	0.333	0.0149(0.0027)	2.35E-08	*MYRF:TMEM258(intronic)*	0.1	Red blood cell traits, White blood cell traits, Platelet traits, Reticulocyte traits, Cholesterol, Triglycerides, Phospholipids, Fatty acids, Glucose, Alkaline phosphatase, Apolipoproteins, Sex hormone-binding globulin, Insulin-like growth factor 1, C-reactive protein, Pulse rate, Height, Heel bone mineral density, Bilirubin levels, Urate levels, Cystatin C levels, Asthma, Diabetes, Endocrine diseases, Hypothyroidism, Cholelithiasis, Gallbladder disorders, Daytime napping, Statin use, Hair/balding pattern
rs10898126	11:83,188,844	G/T	0.3827	−0.0143(0.0025)	1.44E-08	*DLG2(intronic)*	0.44	Long-standing illness, disability or infirmity, Oily fish intake, Overall health rating
rs4843053	15:85,921,559	T/G	0.4861	0.0148(0.0025)	4.55E-09	*AKAP13(intergenic)*	0.26	Height, Appendicular lean mass, Neutrophil percentage of white cells, Mean spheric corpuscular volume, Mean sphered cell volume, Neutrophil count
rs12604076	17:76,773,638	T/C	0.4831	0.0145(0.0025)	3.19E-09	*CYTH1(intronic)*	0.43	Height, Basal metabolic rate, Cardiovascular disease, Hypertension, Atrial fibrillation
rs1788820	18:21,101,944	G/A	0.3479	0.0159(0.0026)	2.11E-09	*C18orf8:NPC1(intronic)*	0.0077	Neutrophil count, White blood cell count, Whole body fat mass, Body mass index, Alcohol intake frequency, Lung function (FVC), Mean sphered cell volume, Granulocyte count, Education
rs10421139	19:2,550,096	T/C	0.3022	0.0172(0.0029)	3.55E-09	*GNG7(intronic)*	0.66	–

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Table 1 (continued)

SNP	Location (chr:pos)	OA/EA	MAF	Beta (SE)	*P* value	Nearest mapped gene (SNP function)	Q_{SNP} *P* value	Previous GWAS associations (*P* value < 5×10^{-8})
rs3865514	19:38,440,967	T/A	0.4304	−0.0160(0.0025)	3.15E-10	*SIPA1L3(intronic)*	0.44	–
rs74172675	19:4,962,970	T/C	0.2793	0.0160(0.0028)	7.85E-09	*UHRF1(downstream)*	0.37	Standing height, Monocyte percentage of white cells, Neutrophil percentage of white cells, Neutrophill percentage, Lymphocyte percentage of white cells, Neutrophil count, Lymphocyte percentage, Neutrophill count, Forced vital capacity (fvc),

SNP, single nucleotide polymorphism; OA, other allele; EA, effect allele; MAF, minor allele frequency

The Q_{SNP} is a Chi-square distributed statistic, the null hypothesis, namely, that the SNP effect is completely mediated by a common pathway (Here was the Faid_cvd factor). A statistically significant Q_{SNP} suggests that the SNP effect is best explained by specific pathways (Faid_cvd components like CAD) that are independent of the shared Faid_cvd factor

Previous GWAS associations were from the Opentargets records

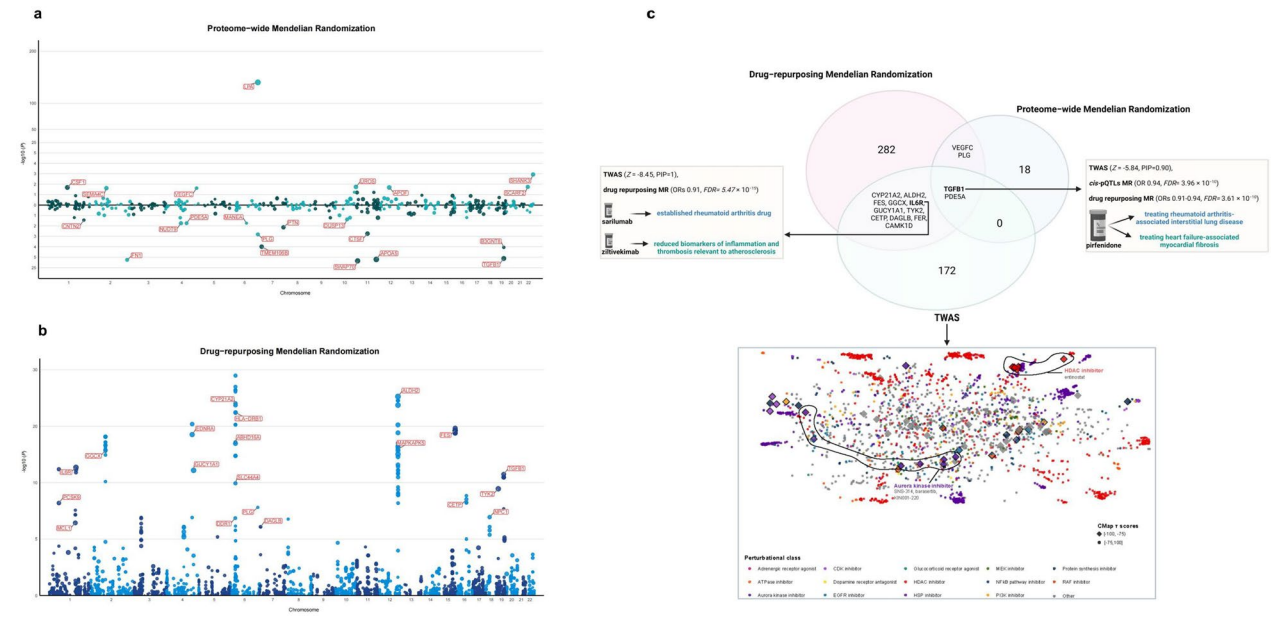


Fig. 4 Proteome-wide and drug-repurposing Mendelian randomization (MR), and computational drug repurposing analyses. **a** proteome-wide MR. Significant (*FDR* < 0.05) associations between circulating proteins and Faid_cvd are highlighted in red and labeled with gene name. Size of points represents absolute effect size (larger points correspond to greater absolute effect size). Points above the origin correspond to proteins where higher circulating levels are associated with increased risk of AD-CVD comorbidity, while those below the origin correspond to proteins where higher circulating levels are associated with decreased risk of AD-CVD comorbidity. **b** Drug-repurposing MR. Significant (*FDR* < 0.05) associations between gene expression levels of actionable drug targets and risk of Faid_cvd are highlighted in red and labeled with gene name. Size of points represents absolute effect size (larger points correspond to greater absolute effect size). Targets are positioned based on the chromosomal location of the protein-coding gene. **c,** The Venn diagram for drug targets prioritized by proteome-wide or drug-repurposing MR and TWAS, individually or jointly. Two representative examples are highlighted: IL6R (joint evidence from drug-repurposing MR and TWAS) with its targeting drugs sarilumab and ziltivekimab, and TGFB1 (joint evidence from TWAS and both MR approaches) with its targeting drug pirfenidone. We used significant TWAS associations ($P < 1.38 \times 10^{-6}$ and posterior inclusion probability > 0.5, and effect directions were consistent across the cross-tissue eQTL weights -sCCA1, sCCA2 and sCCA3) in CMap to identify drugs that could potentially reverse the disease signature. Data are visualized with L1000FWD. Each point represents cell-specific drug-induced signature, and signatures were clustered via *k*-nearest neighbors. Diamond shape represents signatures with CMap τ scores less than -75 (details see **Methods**). Signatures are colored by perturbational drug classes. Contour lines and labels represent frequently identified drug class and associated drugs

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

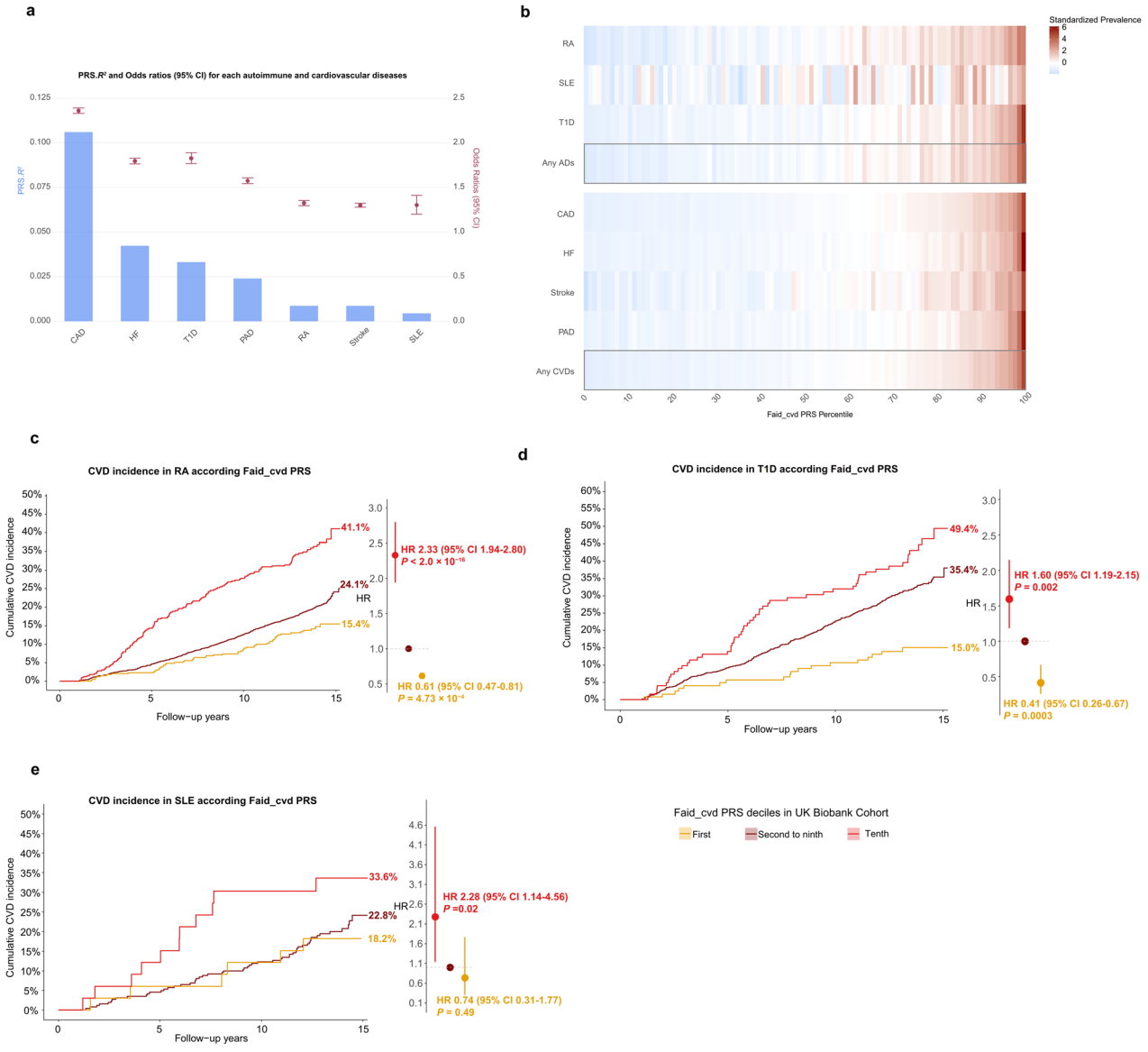


Fig. 5 Polygenic risk score (PRS) of Faid_cvd in the 337,049 participants from UK Biobank. **a** Phenotypic variance (R^2) of individual AD and CVD explained by the Faid_cvd PRS. Odd ratios (95% confident interval [CI]) of PRS with individual AD and CVD, estimated by logistic regression model with adjustments of age, sex, genetic batches, assessment center and the first 10 genetic principal components. **b** Association patterns of AD and CVD prevalence with Faid_cvd PRS percentiles. Disease prevalence was standardized within each disease category using *z*-score transformation, the standardized prevalence are shown in a heatmap with color gradient from blue (lower than mean) through white (mean) to red (higher than mean). **c-e** Cumulative incidence (95% CI) and hazard ratios (95% CI) of Faid_cvd PRS with incident CVD risk among autoimmune patients (RA, T1D, SLE), estimated by cox regression model with adjustments of age, sex, genetic batches, assessment center and the first 10 genetic principal components

autoimmune susceptibility and *ATXN2*-associated metabolic dysregulation may synergistically contribute to cardiovascular disease onset [34]. This gene cluster's coordinated expression effects potentially elucidate the molecular nexus underlying RA/T1D-CVD comorbidity. Moreover, we identified significant local genetic correlations between all three ADs and CAD in the genomic locus region on chromosome 1 (113,418,038–114,664,387), which encompasses *PTPN22*. This gene is widely recognized as the most influential non-major histocompatibility complex (MHC) gene in promoting autoimmunity development, recent evidence also suggests its role in enhancing atherosclerosis progression [45, 46]. While abundant polygenic overlaps underlie AD-CVD pairs, more nuanced information about potential pleiotropic genes, such as the direction and magnitude of their effects, which requires a more general cross-phenotype shared genetic basis to explain the underlying biological mechanism for AD-CVD comorbidity.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

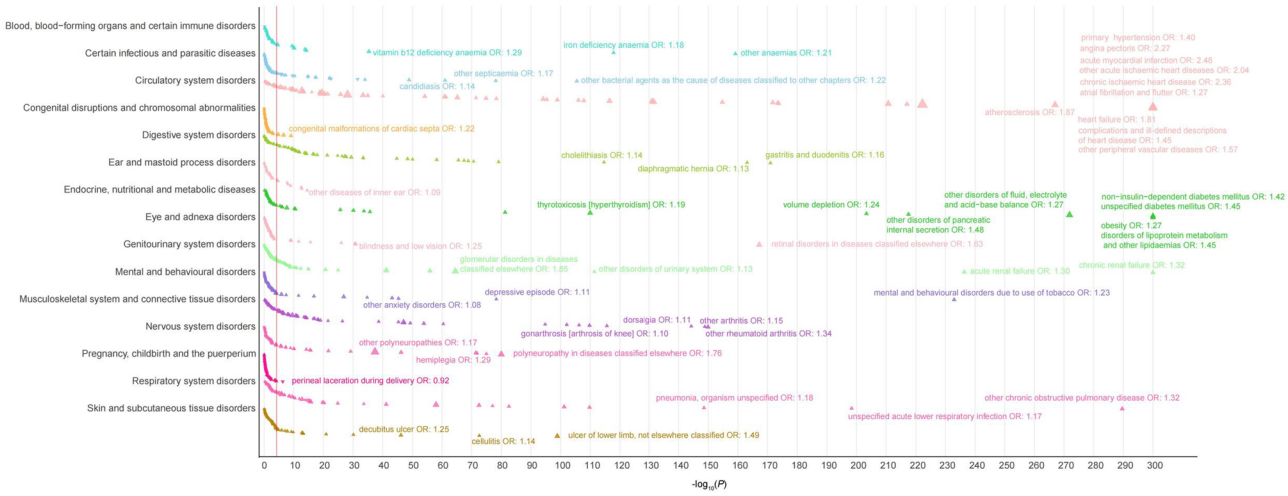


Fig. 6 Phenome-wide associations of Faid_cvd with health outcomes. Where the *x* axis refers to the ICD-10 code category, the *y* axis refers to the significance on a $-\log_{10}$ scale, the vertical light red line denotes phenome-wide significance ($P = 1 \times 10^{-5}$). The direction of the triangle refers to the direction of effect, upward triangle represents positive association, downward represents negative association

Our analysis uncovered several variants that were not detected in the individual GWAS of the three autoimmune diseases and four cardiovascular diseases under this study, exhibiting significant associations with key risk factors and processes implicated in cardiovascular etiology. For example, rs12641981 showed an association with BMI and diabetes mellitus; rs108499 demonstrated pleiotropic associations with multiple lipid-metabolic factors, including phospholipids, triglyceride, and apolipoproteins A1 and B. Our findings of many lead variants associated with these factors, suggesting that the strict management of these factors could potentially modulate AD-CVD comorbidity risk. We also identified several high-confidence genes (such as *PCSK9* and *APOE*) using fine mapping. *PCSK9* exhibits pleiotropic effects, influencing both inflammation and immunity [47]. Silencing of *PCSK9* not only attenuates Oxidized LDL (OxLDL)-induced T cell activation by dendritic cells but also promotes an anti-inflammatory phenotype, reversing OxLDL-mediated polarization towards Th1 and Th17 subsets, this mechanism potentially offers therapeutic benefits for SLE and RA [48, 49], addressing both disease activity and associated complications like CVD. Conditional deletion of the *APOE* gene in adult mice rapidly induces T and B cell responses, leading to germinal center formation and systemic autoimmunity, this autoimmune response promotes atherosclerotic lesion development, potentially elucidating a mechanistic link between atherosclerosis and rheumatological disorders [50]. The Faid_cvd signature, representing the shared genetic basis of all seven specific autoimmunity and cardiovascular phenotypes, includes a subset of prioritized variants and genes previously implicated in AD-CVD comorbidity and related processes. This signature not only illuminates potential pathways connecting risk factors to AD-CVD comorbidity but also identifies promising loci for future investigation, potentially deepening our understanding of the underlying mechanisms.

Our study observed predominant enrichment in several biological pathways including lipid metabolism, immune inflammatory response, cellular signaling, and extracellular matrix remodeling, which are highly aligned with currently established pathophysiological evidence linking between autoimmunity and atherosclerosis [51–55]. Our analysis revealed extracellular matrix (ECM) organization as one of the most significantly enriched pathways. The ECM plays an essential role in vascular endothelium function, regulating endothelial cell migration, invasion, proliferation, and survival [56], these ECM-mediated processes are crucial in the early stages of atherogenesis [56]. The *COL4A1* and *COL4A2* genes (chromosome 13q34) encode the α1 and α2 chains of collagen IV, the major constituent of basement membranes (highly specified ECM). Our study identified the intronic variant rs4773144 as a putative causal variant (posterior probability > 0.95), this variant has been proven to play an important affecting *COL4A1* and *COL4A2* expression, smooth muscle cell survival, and atherosclerotic plaque stability, potentially linking basement membrane composition to atherogenesis [57]. Furthermore, our study demonstrated significant enrichment in pathways related to plasma lipoprotein assembly, remodeling, clearance, and lipid metabolic processes. This finding was supported by the identification of several high-confidence genes including *PCSK9, LPA, PLG, LPL, APOE*, and *LDLR*. These identified genes highlighted the importance of low-density lipoprotein cholesterol (LDL-C) for AD-CVD comorbidity [52],

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

aligning with the well-established causal role of LDL-C in the development and progression of atherosclerotic cardiovascular disease: the accumulation of LDL-C (and other apolipoprotein B-containing lipoproteins) in the subendothelial space leads to its oxidation, triggering inflammation and the recruitment of monocytes into the arterial intima. Then, these monocytes differentiate into macrophages, which then take up oxidized LDL to form foam cells, initiating the development of atherosclerotic plaques [58–60]. Our enrichment analysis highlighted key mediators of inflammatory response, particularly interleukin-6 (IL-6) signaling, in CVD and autoimmunity. Genetic studies have shown that variants affecting IL-6 signaling, which increase IL-6R cleavage and plasma levels, correlate with lifelong CRP levels and cardiovascular risk [61, 62], and phenome-wide association studies have linked IL6R variants not only to atherosclerosis but also to aneurysm formation in multiple arterial beds [63]. Notably, tocilizumab, an IL-6-targeted therapy, has been in use for rheumatoid arthritis for several years, which treatment has shown promise in ameliorating endothelial dysfunction, as evidenced by enhanced flow-mediated dilatation and reduced aortic stiffness measured via pulse wave velocity [64, 65]. The IL-6-targeted therapy could potentially emerge as a therapeutic option to mitigate the elevated cardiovascular risk associated with autoimmune disorders.

In the UK Biobank cohort, we found that the Faid_cvd PRS demonstrated relatively good predictive capability across both autoimmune and cardiovascular diagnoses. However, its explanatory power varied between disease categories. For autoimmune phenotypes, specifically RA and SLE, the PRS explained less than 1% of the phenotypic variance ($R^2 < 0.01$). In contrast, for cardiovascular phenotypes, the PRS consistently explained more than 1% of the variance ($R^2 > 0.01$). This discrepancy in phenotypic variance explanation may due to the fact that the imputed HLA variation for RA and SLE are also essential for polygenic risk prediction [66, 67], our PRS did not include them as it was designed to characterize the general shared genetic basis across autoimmune and cardiovascular disease phenotypes. Further, we assessed PRS-based predictions for developing cardiovascular comorbidity over time and demonstrated statistically significant differences in hazard rates and absolute risks between individuals with autoimmune diseases in the highest PRS deciles compared to second-ninth deciles for cardiovascular outcomes (eg. CAD, HF). This risk prediction could probably be further improved by including other risk factors such as family history and clinical/phenotypic variables [68]. Our findings have potentially significant clinical implications: based on the PRS (or, ideally, a combination of genetic and clinical risk factors), could be helpful to identify individuals with autoimmune diseases who are at higher risk of developing cardiovascular complications, leading to earlier diagnosis and timely initiation of preventive or therapeutic interventions. However, it is crucial to emphasize that the practical application of these findings requires thorough validation in clinical settings through future studies [69].

Our phenome-wide association analysis revealed that the genetic liability for AD-CVD comorbidity is associated with numerous medical phenotypes, including polyneuropathy, chronic renal failure, and depression. These associations underscore the complex interplay between autoimmune diseases, cardiovascular conditions, and other systemic complications. Polyneuropathy, for instance, is a known complication in autoimmune diseases such as RA and SLE. Its etiology in these conditions is multifaceted, involving compression due to joint deformation or swelling, vasculitis, and drug toxicity [70]. Moreover, neuropathy is also the most prevalent complication among diabetes patients, affecting over 50% of diabetes patients over time [71]. Similarly, chronic kidney disease frequently co-occurs with RA, SLE, and T1D [72–74]. Patients with these conditions often present with more traditional cardiovascular risk factors, consequently facing an elevated risk of CVD and mortality. Depression, another phenotype associated with AD-CVD genetic liability, demonstrates genome-wide pleiotropy with autoimmune diseases. Research has shown significantly higher polygenic risk scores (PRSs) for depression in individuals with rheumatoid arthritis and type 1 diabetes compared to controls [75]. Furthermore, a separate study has revealed a significant genetic overlap between CVD and major depressive disorder, attributed to shared inflammatory, metabolic, and psychosocial or lifestyle risk factors [76].These investigations, together with our results, collectively suggest the genetic liability for AD-CVD comorbidity extends beyond the primary autoimmune and cardiovascular systems, influencing a broader spectrum of health outcomes, which indicating that timely monitoring of complications or targeted treatment is more necessary in individuals with high genetic susceptibility to AD-CVD.

Our study identifies promising drug-repurposing opportunities, providing human genetic evidence to motivate further mechanistic studies and clinical trials. These findings highlight potential therapeutic targets that may benefit patients with both AD and CVD. For instance, our prioritization of *SWAP70* offers human genetic evidence for developing a novel drug that could potentially address both AD and CVD. Furthermore, our findings reinforce the role of key pathways in AD-CVD comorbidity, including LDL metabolic processes (e.g., *PCSK9*), inflammation (e.g., *IL6R*), and epithelial cell proliferation (e.g., *EDNRA*). While future mechanistic evaluation of these targets is warranted, the genetic evidence

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

for these pathways increases the likelihood of successful clinical translation [77, 78].

Our study has certain limitations that warrant acknowledgment. Firstly, the univariate GWAS data used as input for the multivariate GWAS analysis (genomic SEM) is derived from populations of European ancestry. Future validations using large-scale GWAS meta-analyses incorporating multiple populations are necessary to ensure broader applicability of our findings. Secondly, we utilized UKB participants as the target sample for constructing the Faid_cvd PRS. While there is indeed sample overlap between the GWAS summary statistics of CVD traits and the UKB, our comparison of R^2 (phenotypic variance) explained by the Faid_cvd PRS for individual AD and CVD traits is presented on a relative rather than absolute scale, which could mitigate concerns about the impact of sample overlap on PRS. Thirdly, our analysis is limited to common variants on autosomal chromosomes, the importance of rare variants has been recognized in autoimmune diseases [79–81], future studies investigating the role of rare variants in this context are warranted to provide a more comprehensive genetic landscape of AD-CVD comorbidity.

Conclusions

In summary, through this multivariate genome-wide association study, we characterized the shared genetic architecture underlying autoimmune diseases and cardiovascular diseases at both locus and polygenic levels. Our findings have important implications for early risk stratification, drug development and therapeutic repositioning in managing the coexisting morbidity of autoimmune and cardiovascular diseases.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12933-025-03041-8.

Polygenic risk scoring file (variants, effect alleles, weights)

Supplementary Methods & Supplementary Figures

Supplementary Tables

Acknowledgements

We want to acknowledge the participants and investigators of the many GWAS studies used in this research without whom this effort would not be possible. This study used the UK Biobank data under the approved project with application number 62663.

Author contributions

JZ, DQY designed the study. JZ conducted the main analyses and drafted the manuscript. JZ, XYY contributed to interpretation of results. JZ, XYY critically revised the manuscript. All authors reviewed and approved the final version of the manuscript, DQY had full access to the data in the study and accept responsibility for its submission for publication.

Funding

This work was supported by National Natural Science Foundation of China (Grant no. 82073652); Anhui Province Clinical Medical Research Transformation Project (NO.202304295107020041; NO.202304295107020048); and Chinese Research Funds of Joint Research Center for Occupational Medicine and Health of IHM(OMH-2023–01;OMH-2023–08). The Science and Technology Development Fund of Pudong New Area (PKJ2023-Y36); Young Medical Talents Training Program of Pudong Health Bureau of Shanghai (PWRq2022-01). The funders of the study had no role in study design, data collection, analysis, decision to publish, or manuscript preparation.

Data availability

Links to download publicly available published GWAS summary statistics data used as inputs in this study are listed in Supplementary Table S1. Researchers can request access to the UKB data resources at https://www.ukbiobank.ac.uk/enable-your-research/apply-for-access. GTEx weights for FUSION analyses are available at http://gusevlab.org/projects/fusion/. Single-cell gene expression data from the Tabula Muris study are available at https://tabula-muris.ds.czbiohub.org/. Summary statistics for GWAS of the AD-CVD latent factors are available via figshare at https://doi.org/10.6084/m9.figshare.27129813https://figshare.com/s/831a1e8744dafcfd9aca. All other data supporting the findings of this study are available from the corresponding author upon reasonable request.

Code availability

The software used in this study is publicly available and accessed without restriction. R package 'GenomicSEM' v.0.0.5 is available at https://github.com/GenomicSEM/GenomicSEM. PRS was constructed using PRSice-2, is available at https://choishingwan.github.io/PRSice/. For implementing Mendelian randomization analyses, R package 'TwoSampleMR' v.0.5.6 and 'MendelianRandomization' v.0.7.0 are publicly available. FUSION Pipeline v.1.4.2 for TWAS analysis is available at http://gusevlab.org/projects/fusion/. Python package FOCUS v.0.6.10 for FOCUS fine mapping for FUSION is available at http://github.com/bogdanlab/focus/. Python package CELLECT v.1.3.0 for single-cell enrichment analyses is available at https://github.com/perslab/CELLECT. R package 'echolocatoR' v.2.0.3 for fine mapping using SuSIE and FINEMAP is available at https://github.com/RajLabMSSM/echolocatoR. MendelVar is available at https://mendelvar.mrcieu.ac.uk/. GeneMANIA is is available at https://genemania.org/ FUMA and MAGMA v.1.4.0 are available at https://fuma.ctglab.nl/. R v.4.1.3 and Python v.3.8 were used to format data for analyses. Software to construct and browse Enrichment Map is freely available (http://www.baderlab.org/Software/EnrichmentMap). All scripts are uploaded in the GitHub (https://github.com/jiezhang-cn/AD_CVD_Genetic_Architecture), and available from the corresponding authors on reasonable request.

Declarations

Conflict of interests

The authors declare no competing interests.

Author details

[1]School of Public Health, Anhui University of Science and Technology, Hefei 231131, Anhui, China
[2]Department of Epidemiology, School of Public Health, Anhui Medical University, Hefei 230032, Anhui, China
[3]Inflammation and Immune Mediated Diseases Laboratory of Anhui Province, Hefei 230032, Anhui, China
[4]Fuyang Affiliated Hospital of Anhui Medical University, Fuyang 236307, Anhui, China
[5]Department of Oncology, School of Medicine, Shanghai East Hospital, Tongji University, Shanghai, China
[6]Department of Epidemiology, School of Public Health, Lianyungang Medical-Education Innovation andResearch Center, Nanjing Medical University, Nanjing 211166, Jiangsu, China
[7]Joint Research Center of Occupational Medicine and Health, Institute of Grand Health, Hefei Comprehensive National Science Center, Hefei 231131, Anhui, China

Received: 22 September 2025 / Accepted: 10 December 2025
Published online: 11 February 2026

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

References

1. Pisetsky DS. Pathogenesis of autoimmune disease. Nat Rev Nephrol. 2023;19:509–24. https://doi.org/10.1038/s41581-023-00720-1.
2. Conrad N, Verbeke G, Molenberghs G, et al. Autoimmune diseases and cardiovascular risk: a population-based study on 19 autoimmune diseases and 12 cardiovascular diseases in 22 million individuals in the UK. Lancet. 2022;400(10354):733–43. https://doi.org/10.1016/S0140-6736(22)01349-6.
3. Cho JH, Feldman M. Heterogeneity of autoimmune diseases: pathophysiologic insights from genetics and implications for new therapies. Nat Med. 2015;21(7):730–8. https://doi.org/10.1038/nm.3897.
4. Harroud A, Hafler DA. Common genetic factors among autoimmune diseases. Science. 2023;380:485–90. https://doi.org/10.1126/science.adg2992.
5. Kathiresan S, Srivastava D. Genetics of human cardiovascular disease. Cell. 2012;148(6):1242–57. https://doi.org/10.1016/j.cell.2012.03.001.
6. Guo Y, Chung W, Shan Z, Zhu Z, Costenbader KH, Liang L. Genome-wide assessment of shared genetic architecture between rheumatoid arthritis and cardiovascular diseases. J Am Heart Assoc. 2023. https://doi.org/10.1161/JAHA.123.030211.
7. Szymczak F, et al. Gene expression signatures of target tissues in type 1 diabetes, lupus erythematosus, multiple sclerosis, and rheumatoid arthritis. Sci Adv. 2021;7:eabf7600. https://doi.org/10.1126/sciadv.abd7600.
8. Grotzinger AD, Rhemtulla M, de Vlaming R, et al. Genomic structural equation modelling provides insights into the multivariate genetic architecture of complex traits. Nat Hum Behav. 2019;3(5):513–25. https://doi.org/10.1038/s41562-019-0566-x.
9. Ishigaki K, Sakaue S, Terao C, et al. Multi-ancestry genome-wide association analyses identify novel genetic mechanisms in rheumatoid arthritis. Nat Genet. 2022;54:1640–51. https://doi.org/10.1038/s41588-022-01213-w.
10. Chiou J, Geusz RJ, Okino ML, et al. Interpreting type 1 diabetes risk with genetics and single-cell epigenomics. Nature. 2021;594:398–402. https://doi.org/10.1038/s41586-021-03552-w.
11. Bentham J, Morris D, Cunninghame Graham D, et al. Genetic association analyses implicate aberrant regulation of innate and adaptive immunity genes in the pathogenesis of systemic lupus erythematosus. Nat Genet. 2015;47:1457–64. https://doi.org/10.1038/ng.3434.
12. Aragam KG, Jiang T, Goel A, et al. Discovery and systematic characterization of risk variants and genes for coronary artery disease in over a million participants. Nat Genet. 2022;54:1803–15. https://doi.org/10.1038/s41588-022-01233-6.
13. Shah S, Henry A, Roselli C, et al. Genome-wide association and Mendelian randomisation analysis provide insights into the pathogenesis of heart failure. Nat Commun. 2020;11:163. https://doi.org/10.1038/s41467-019-13690-5.
14. Mishra A, Malik R, Hachiya T, et al. Stroke genetics informs drug discovery and risk prediction across ancestries. Nature. 2022;611:115–23. https://doi.org/10.1038/s41586-022-05165-3.
15. van Zuydam NR, Stiby A, Abdalla M, et al. Genome-wide association study of peripheral artery disease. Circ Genom Precis Med. 2021;14(5):e002862. https://doi.org/10.1161/CIRCGEN.119.002862.
16. Bulik-Sullivan B, Loh PR, Finucane H, et al. LD score regression distinguishes confounding from polygenicity in genome-wide association studies. Nat Genet. 2015;47:291–5.
17. Werme J, van der Sluis S, Posthuma D, et al. An integrated framework for local genetic correlation analysis. Nat Genet. 2022;54:274–82. https://doi.org/10.1038/s41588-022-01017-y.
18. Watanabe K, Taskesen E, van Bochoven A, Posthuma D. Functional mapping and annotation of genetic associations with FUMA. Nat Commun. 2017;8:1826.
19. Yang J, et al. Conditional and joint multiple-SNP analysis of GWAS summary statistics identifies additional variants influencing complex traits. Nat Genet. 2012;44:369–75.
20. Zou Y, Carbonetto P, Wang G, Stephens M. Fine-mapping from summary data with the "Sum of Single Effects" model. PLoS Genet. 2022;18(7):e1010299. https://doi.org/10.1371/journal.pgen.1010299.
21. Benner C, Spencer CC, Havulinna AS, Salomaa V, Ripatti S, Pirinen M. FINEMAP: efficient variable selection using summary data from genome-wide association studies. Bioinformatics. 2016;32(10):1493–501. https://doi.org/10.1093/bioinformatics/btw018.
22. Finucane HK, et al. Partitioning heritability by functional annotation using genome-wide association summary statistics. Nat Genet. 2015;47:1228–35.
23. Finucane HK, et al. Heritability enrichment of specifically expressed genes identifies disease-relevant tissues and cell types. Nat Genet. 2018;50:621–9.
24. de Leeuw CA, Mooij JM, Heskes T, Posthuma D. MAGMA: generalized gene-set analysis of GWAS data. PLoS Comput Biol. 2015;11:e1004219.
25. Sobczyk MK, Gaunt TR, Paternoster L. MendelVar: gene prioritization at GWAS loci using phenotypic enrichment of Mendelian disease genes. Bioinformatics. 2021;37:1–8.
26. Gusev A, et al. Integrative approaches for large-scale transcriptome-wide association studies. Nat Genet. 2016;48:245–52.
27. Feng H, Mancuso N, Gusev A, et al. Leveraging expression from multiple tissues using sparse canonical correlation analysis and aggregate tests improves the power of transcriptome-wide association studies. PLoS Genet. 2021;17(4):e1008973.
28. Mancuso N, et al. Probabilistic fine-mapping of transcriptome-wide association studies. Nat Genet. 2019;51:675–82.
29. Lamb J, Crawford ED, Peck D, et al. The connectivity map: using gene-expression signatures to connect small molecules, genes, and disease. Science. 2006;313(5795):1929–35. https://doi.org/10.1126/science.1132939.
30. Zheng J, et al. Phenome-wide Mendelian randomization mapping the influence of the plasma proteome on complex diseases. Nat Genet. 2020;52:1122–31.
31. Gaziano L, et al. Actionable druggable genome-wide Mendelian randomization identifies repurposing opportunities for COVID-19. Nat Med. 2021;27:668–76.
32. Bycroft, Freeman C, Petkova D, et al. The UK Biobank resource with deep phenotyping and genomic data. Nature. 2018;562:203–9.
33. Choi SW, O'Reilly PF. PRSice-2: polygenic risk score software for biobank-scale data. Gigascience. 2019;8:giz082. https://doi.org/10.1093/gigascience/giz082.
34. Orrù V, Steri M, Sole G, et al. Genetic variants regulating immune cell levels in health and disease. Cell. 2013;155(1):242–56. https://doi.org/10.1016/j.cell.2013.08.041.
35. Flister MJ, Hoffman MJ, Lemke A, et al. SH2B3 is a genetic determinant of cardiac inflammation and fibrosis. Circ Cardiovasc Genet. 2015;8(2):294–304. https://doi.org/10.1161/CIRCGENETICS.114.000527.
36. Benn M, Nordestgaard BG, Grande P, Schnohr P, Tybjaerg-Hansen A. PCSK9 R46L, low-density lipoprotein cholesterol levels, and risk of ischemic heart disease: 3 independent studies and meta-analyses. J Am Coll Cardiol. 2010;55(25):2833–42. https://doi.org/10.1016/j.jacc.2010.02.044.
37. Nelson CP, Lai FY, Nath M, et al. Genetic assessment of potential long-term on-target side effects of PCSK9 (proprotein convertase subtilisin/kexin type 9) inhibitors. Circ Genom Precis Med. 2019;12(1):e002196. https://doi.org/10.1161/CIRCGEN.118.002196.
38. Chung CP, Solus JF, Oeser A, et al. Genetic variation and coronary atherosclerosis in patients with systemic lupus erythematosus. Lupus. 2014;23(9):876–80. https://doi.org/10.1177/0961203314530019.
39. Lewis GA, Dodd S, Clayton D, et al. Pirfenidone in heart failure with preserved ejection fraction: a randomized phase 2 trial. Nat Med. 2021;27:1477–82. https://doi.org/10.1038/s41591-021-01452-0.
40. Ridker PM, Rane M. Interleukin-6 signaling and anti-interleukin-6 therapeutics in cardiovascular disease. Circ Res. 2021;128(11):1728–46. https://doi.org/10.1161/CIRCRESAHA.121.3190774(10):1638-1647.
41. Pedro Ferreira J, Pitt B, Zannad F. Histone deacetylase inhibitors for cardiovascular conditions and healthy longevity. Lancet Healthy Longev. 2021;2(6):e371–9. https://doi.org/10.1016/S2666-7568(21)00061-1.
42. Qiao J, Chang M, Chen M, Zhao Y, Hao J, Zhang P, et al. Genetic analyses identify shared genetic components related to autoimmune and cardiovascular diseases. medRxiv. 2024. https://doi.org/10.1101/2024.09.01.24310190.
43. Yuan S, Carter P, Mason AM, Yang F, Burgess S, Larsson SC. Genetic liability to rheumatoid arthritis in relation to coronary artery disease and stroke risk. Arthritis Rheumatol. 2022;74(10):1638–47. https://doi.org/10.1002/art.42239.
44. Adebekun J, Nadig A, Saarah P, Asgari S, Kachuri L, Alagpulinsa DA. Genetic relations between type 1 diabetes, coronary artery disease and leukocyte counts. Diabetologia. 2024;67(11):2518–29. https://doi.org/10.1007/s00125-024-06247-9.
45. Tizaoui K, Terrazzino S, Cargnin S, et al. The role of PTPN22 in the pathogenesis of autoimmune diseases: a comprehensive review. Semin Arthritis Rheum. 2021;51(3):513–22. https://doi.org/10.1016/j.semarthrit.2021.03.004.
46. Dai X, James RG, Habib T, et al. A disease-associated PTPN22 variant promotes systemic autoimmunity in murine models. J Clin Invest. 2013;123(5):2024–36. https://doi.org/10.1172/JCI66963.
47. Frostegård J. The role of PCSK9 in inflammation, immunity, and autoimmune diseases. Expert Rev Clin Immunol. 2022;18(1):67–74. https://doi.org/10.1080/1744666X.2022.2017281.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

48. Sanchez-Perez H, Quevedo-Abeledo JC, Tejera-Segura B, et al. Proprotein convertase subtilisin/kexin type 9 is related to disease activity and damage in patients with systemic erythematosus lupus. Ther Adv Musculoskelet Dis. 2020;12:1759720X20975904.
49. Frostegard J, Ahmed S, Hafstrom I, et al. Low levels of PCSK9 are associated with remission in patients with rheumatoid arthritis treated with anti-TNF-alpha: potential underlying mechanisms. Arthritis Res Ther. 2021;23(1):32.
50. Centa M, Prokopec KE, Garimella MG, et al. Acute loss of apolipoprotein E triggers an autoimmune response that accelerates atherosclerosis. Arterioscler Thromb Vasc Biol. 2018;38(8):e145–58. https://doi.org/10.1161/ATVBAHA.118.310802.
51. Porsch F, Binder CJ. Autoimmune diseases and atherosclerotic cardiovascular disease. Nat Rev Cardiol. 2024. https://doi.org/10.1038/s41569-024-01045-7.
52. Kerns S, Owen KA, Daamen A, Kain J, Grammer AC, Lipsky PE. Genetic association with autoimmune diseases identifies molecular mechanisms of coronary artery disease. iScience. 2024;27(9):110715. https://doi.org/10.1016/j.isci.2024.110715.
53. Ambler WG, Kaplan MJ. Vascular damage in systemic lupus erythematosus. Nat Rev Nephrol. 2024;20:251–65. https://doi.org/10.1038/s41581-023-00797-8.
54. Weber BN, Giles JT, Liao KP. Shared inflammatory pathways of rheumatoid arthritis and atherosclerotic cardiovascular disease. Nat Rev Rheumatol. 2023;19:417–28. https://doi.org/10.1038/s41584-023-00969-7.
55. Vergès B. Cardiovascular disease in type 1 diabetes, an underestimated danger: epidemiological and pathophysiological data. Atherosclerosis. 2024;394:117158. https://doi.org/10.1016/j.atherosclerosis.2023.06.005.
56. Davis GE, Senger DR. Endothelial extracellular matrix: biosynthesis, remodeling, and functions during vascular morphogenesis and neovessel stabilization. Circ Res. 2005;97(11):1093–107. https://doi.org/10.1161/01.RES.0000191547.64391.e3.
57. Yang W, Ng FL, Chan K, et al. Coronary-heart-disease-associated genetic variant at the COL4A1/COL4A2 locus affects COL4A1/COL4A2 expression, vascular cell survival, atherosclerotic plaque stability and risk of myocardial infarction. PLoS Genet. 2016;12(7):e1006127. https://doi.org/10.1371/journal.pgen.1006127.
58. Borén J, et al. Low-density lipoproteins cause atherosclerotic cardiovascular disease: pathophysiological, genetic, and therapeutic insights: a consensus statement from the European Atherosclerosis Society Consensus Panel. Eur Heart J. 2020;41:2313–30.
59. Moore KJ, Sheedy FJ, Fisher EA. Macrophages in atherosclerosis: a dynamic balance. Nat Rev Immunol. 2013;13:709–21.
60. Ronda N, Zimetti F, Adorni MP, Palumbo M, Karpouzas GA, Bernini F. Role of lipoprotein levels and function in atherosclerosis associated with autoimmune rheumatic diseases. Rheum Dis Clin North Am. 2023;49(1):151–63. https://doi.org/10.1016/j.rdc.2022.07.006.
61. Sarwar N, Butterworth AS, Freitag DF, Gregson J, Willeit P, Gorman DN, et al. IL6R genetics consortium emerging risk factors collaboration. Interleukin-6 receptor pathways in coronary heart disease: a collaborative meta-analysis of 82 studies. Lancet. 2012;379:1205–13. https://doi.org/10.1016/S0140-6736(11)61931-4.
62. Swerdlow DI, Holmes MV, Kuchenbaecker KB, Engmann JE, Shah T, Sofat R, et al. Interleukin-6 receptor mendelian randomisation analysis (IL6R MR) consortium. The interleukin-6 receptor as a target for prevention of coronary heart disease: a Mendelian randomisation analysis. Lancet. 2012;379:1214–24. https://doi.org/10.1016/S0140-6736(12)60110-X.
63. Cai T, Zhang Y, Ho YL, Link N, Sun J, Huang J, et al. Association of interleukin 6 receptor variant with cardiovascular disease effects of interleukin 6 receptor blocking therapy: a phenome-wide association study. JAMA Cardiol. 2018;3:849–57. https://doi.org/10.1001/jamacardio.2018.2287.
64. Bacchiega BC, et al. Interleukin 6 inhibition and coronary artery disease in a high-risk population: a prospective community-based clinical study. J Am Heart Assoc. 2017;6:e005038.
65. Protogerou AD, et al. A pilot study of endothelial dysfunction and aortic stiffness after interleukin-6 receptor inhibition in rheumatoid arthritis. Atherosclerosis. 2011;219:734–6.
66. Yarwood A, Han B, Raychaudhuri S, et al. A weighted genetic risk score using all known susceptibility variants to estimate rheumatoid arthritis risk. Ann Rheum Dis. 2015;74(1):170–6. https://doi.org/10.1136/annrheumdis-2013-204133.
67. Knevel R, le Cessie S, Terao CC, et al. Using genetics to prioritize diagnoses for rheumatology outpatients with inflammatory arthritis. Sci Transl Med. 2020;12(545):eaay1548. https://doi.org/10.1126/scitranslmed.aay1548.
68. O'Sullivan JW, Raghavan S, Marquez-Luna C, et al. Polygenic risk scores for cardiovascular disease: a scientific statement from the american heart association. Circulation. 2022;146(8):e93–118. https://doi.org/10.1161/CIR.0000000000001077.
69. Lewis CM, Vassos E. Polygenic risk scores: from research tools to clinical instruments. Genome Med. 2020;12:44. https://doi.org/10.1186/s13073-020-00742-5.
70. Vallat JM, Rabin M, Magy L. Peripheral neuropathies in rheumatic disease—a guide to diagnosis. Nat Rev Rheumatol. 2012;8:599–609. https://doi.org/10.1038/nrrheum.2012.138.
71. Feldman EL, Callaghan BC, Pop-Busui R, et al. Diabetic neuropathy. Nat Rev Dis Primers. 2019;5:41. https://doi.org/10.1038/s41572-019-0092-1.
72. Wong CY, Ma BMY, Zhang D, Cheung W, Chan TM, Yap DYH. Cardiovascular risk factors and complications in patients with systemic lupus erythematosus with and without nephritis: a systematic review and meta-analysis. Lupus Sci Med. 2024;11(1):e001152. https://doi.org/10.1136/lupus-2024-001152.
73. van Sijl AM, van den Oever IA, Peters MJ, Boers M, Dijkmans BA, van Halm VP, et al. Subclinical renal dysfunction is independently associated with cardiovascular events in rheumatoid arthritis: the CARRE Study. Ann Rheum Dis. 2012;71:341–4. https://doi.org/10.1136/annrheumdis-2011-200051.
74. Heerspink HJ, Cherney DZ, Groop PH, et al. People with type 1 diabetes and chronic kidney disease urgently need new therapies: a call for action. Lancet Diabetes Endocrinol. 2023;11(8):536–40. https://doi.org/10.1016/S2213-8587(23)00168-7ne.0136508.
75. Glanville KP, Coleman JRI, O'Reilly PF, Galloway J, Lewis CM. Investigating pleiotropy between depression and autoimmune diseases using the UK Biobank. Biol Psychiatry Glob Open Sci. 2021;1(1):48–58. https://doi.org/10.1016/j.bpsgos.2021.03.002.
76. Bergstedt J, Pasman JA, Ma Z, et al. Distinct biological signature and modifiable risk factors underlie the comorbidity between major depressive disorder and cardiovascular disease. Nat Cardiovasc Res. 2024;3:754–69. https://doi.org/10.1038/s44161-024-00488-y.
77. King EA, Davis JW, Degner JF. Are drug targets with genetic support twice as likely to be approved? Revised estimates of the impact of genetic support for drug mechanisms on the probability of drug approval. PLoS Genet. 2019;15:e1008489.
78. Nelson MR, et al. The support of human genetic evidence for approved drug indications. Nat Genet. 2015;47:856–60.
79. Kronzer VL, Sparks JA, Raychaudhuri S, et al. Low-frequency and rare genetic variants associated with rheumatoid arthritis risk. Nat Rev Rheumatol. 2024;20:290–300. https://doi.org/10.1038/s41584-024-01096-7.
80. Jiang SH, Athanasopoulos V, Ellyard JI, et al. Functional rare and low frequency variants in BLK and BANK1 contribute to human lupus. Nat Commun. 2019;10:2201. https://doi.org/10.1038/s41467-019-10242-9.
81. Forgetta V, Manousaki D, Istomine R, et al. Rare genetic variants of large effect influence risk of type 1 diabetes. Diabetes. 2020;69(4):784–95. https://doi.org/10.2337/db19-0831.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH ("Springer Nature").

Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users ("Users"), for small-scale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use ("Terms"). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.

These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.

We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.

While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval , sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.

These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.

To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.

Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.

If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at

onlineservice@springernature.com